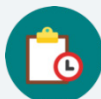
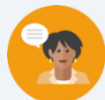


3.8 Angle Relationship Proofs – Part 1

Lesson Introduction				
 Benchmarks	MA.912.GR.1.1: Prove relationships and theorems about lines and angles. Solve mathematical and real-world problems involving postulates, relationships, and theorems of lines and angles. MA.912.LT.4.10 Judge the validity of arguments and give counterexamples to disprove statements.		 Learning Target	I can prove relationships using parallel lines and their transversals.
 Benchmark Coherence	Building On		Working Toward	
	MA.8.GR.1.4 Solve mathematical problems involving the relationships between supplementary, complementary, vertical, or adjacent angles.		N/A	
 Essential Questions	- How can you prove relationships using parallel lines and their transversals? - What are the similarities and differences between two-column, paragraph, and flow chart proofs?			
 Lesson Narrative	This lesson uses the angle pair relationships of parallel lines intersected by transversals to complete formal two-column, flow chart, and paragraph proofs. Students will see that some proofs have multiple paths, all of which are correct if there is no contradiction of information and each statement in the proof is built on a given fact, a justification from a given fact, or previous statement. The lesson includes scaffolded collaboration and explicit guided instruction on the process.			
 Component Summary	3.8.1 Warm-Up (~5 min.)	Would You Rather with two job scenarios (<i>SE p. 164</i>)	3.8.3 Guided Instruction (15 min.)	Complete a flow-chart proof and a paragraph proof involving parallel lines and angle relationships. (<i>SE p. 167</i>)
	3.8.2 Collaboration (20-25 min.)	Order given steps in proofs involving parallel lines and angle relationships (<i>SE p. 165</i>)	3.8.4 Wrap-Up (~5 min.)	Reflect on confidence writing proofs (<i>SE p. 169</i>)
 Resources	Glossary		Additional Preparation	
	<u>Key Terms:</u> N/A <u>Additional Terms:</u> - Parallel Lines - Transversal - Congruent - Supplementary		N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may not realize they can complete proofs in a different order and still be correct.	Students sometimes lack confidence when writing proofs. Students will need to be continually encouraged to write down everything they know about a given situation and everything they can deduct from what they know. Even if they cannot complete the proof in its entirety, they should be encouraged to start with this as their proof-writing skills continue to develop.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Notice and Wonder 3.8.2 Collaboration question 1 provides the opportunity for students to discover what information given is contradictory. Before moving on to questions 1a, 1b, and 1c, consider using this routine for students to make observations about what they know is true about the image.	MA.K12.MTR.1.1: Participate in Effortful Learning Students will be working to complete proofs in various forms. Students who participate in effortful learning, both individually and with others, analyze the problems in a way that makes sense given the task and build perseverance by modifying methods as needed while solving a challenging task. Use the components in this lesson to develop students' ability to analyze and problem-solve.
 Differentiate Instruction	Consider providing some proper placement of statements and/or reasons for the proof in 3.8.2 Collaboration, question 2. Providing this extra scaffolding can offer a different entry point with students who are struggling with ordering the statements and reasons.	
Lesson Notes:		

3.8.1 Warm-Up: Would You Rather?

Component Overview: Students compare jobs at two restaurants to determine which one they would prefer to have.



3.8.1 Warm-Up: Would You Rather?

- The meals at two restaurants range from \$8.00 to \$25.00. Servers are paid based on the following.

Restaurant A	Restaurant B
\$18 per hour	\$10.50 per hour
No tipping allowed	Tipping encouraged

Would you rather work as a server at Restaurant A or at Restaurant B? Justify your choice.

Answers will vary. I would choose Restaurant A because the meals may be less expensive and require less work with serving tables. I would choose Restaurant B because it could be a place where the meals are more expensive, and I have a good personality to where I can provide great service and be nice to customers and make a lot more in tips.



Think-Pair-Share

Consider implementing the Think-Pair-Share instructional routine for 3.8.1 Warm-Up. Students will make better sense of thinking and reasoning with the opportunity to independently process, then refine answers with a partner, before the teacher elicits more precise responses as a whole group.



Challenge students to find a rationale that would make both choices equally appealing.

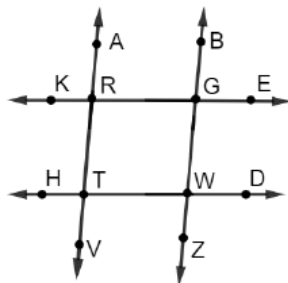
3.8.2 Collaboration: Parallel Lines

Component Overview: Students examine the given information for a figure to determine if any of the information contradicts itself. Students will then place the steps of a two-column proof in an order that justifies a statement to be proven. Then, the students complete the same proof in an alternative way to show that proofs can follow different correct paths.



3.8.2 Collaboration: Parallel Lines

1. Contradictory information is given for the figure.
 $\overleftrightarrow{KE} \parallel \overleftrightarrow{HD}$, $\angle KRA \cong \angle BGE$, $\angle TRG$ is a supplement of $\angle RGW$,
 and $m\angle DWZ = 87^\circ$.



- a. What piece of information is contradictory?
Answers will vary. $m\angle DWZ = 87^\circ$
- b. How do you know it is contradictory?
Answers will vary. If $\angle KRA \cong \angle BGE$ and $\angle TRG$ is a supplement of $\angle RGW$, then both transversals would have to be parallel and intersect the other two parallel lines at 90° angles. But, since the $m\angle DWZ = 87^\circ$, this contradicts the angle being equal to 90° .
- c. Share your contradiction with your partner. Discuss any differences.
- d. Create four different pieces of information about this figure using $\overleftrightarrow{AV} \parallel \overleftrightarrow{BZ}$. One of the pieces of information should be contradictory.
Answers will vary.

1.	$\angle HTV \cong \angle TWZ$
2.	$\angle RTW$ is a supplement of $\angle GWT$
3.	$\angle HTR \cong \angle KRT$ (contradictory statement)
4.	$\angle KRT \cong \angle RGW$



Notice and Wonder

Question 1 provides the opportunity for students to discover what information given is contradictory. Before moving on to questions 1a, 1b, and 1c, consider using this routine to provide the opportunity for students to make observations about what they know is true from the given information, before engaging with the questions.



This collaboration is scaffolded to allow students to engage with it with their partner, rather than being guided through by the teacher. There will be an opportunity to summarize and synthesize what is discovered in the next component. Students may revise their thinking throughout the exploration as they engage with each new reflection. Observe and listen as students work with their partners. Make note of student ideas that you can ask them to share in the whole-group 3.8.3 Guided Instruction that follows. Additional scaffolding questions are provided, if needed, but this should be a primarily student-directed collaboration.

3.8.2 Collaboration: Parallel Lines (continued)

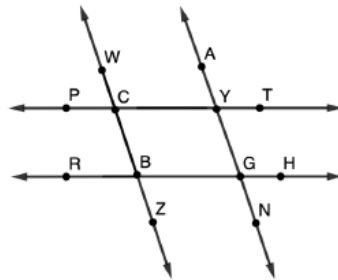
- e. Trade information with a partner and see if your partner can find the contradiction. Write their contradiction in the table.

Partner's Contradiction	Partner's Initials
<i>Answers will vary.</i>	

2. Terrence was organizing a card sort for the proof, but his cards fell on the floor and are now out of order.

Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and
 $\angle CBG \cong \angle CYG$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



Statement	Reason
A. $m\angle CBG = m\angle CYG$	K. Definition of Supplementary Angles
B. $\angle CBG$ and $\angle BGY$ are supplementary	L. Substitution Property of Equality
C. $\angle CBG \cong \angle CYG$	M. If two lines are intersected by a transversal so that consecutive interior angles are supplementary, then the lines are parallel.
D. $\angle CYG$ and $\angle BGY$ are supplementary	N. Definition of Congruence
E. $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$	P. Given
F. $m\angle CBG + m\angle BGY = 180$	R. If two parallel lines are intersected by a transversal, then the consecutive interior angles are supplementary.
G. $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$	S. Definition of Supplementary Angles
H. $m\angle CYG + m\angle BGY = 180$	T. Given



Notice 3.8.2 Collaboration, question 1 provides a variety of entry points for learners to engage with the content. Question 1a provides visual cues, question 1b offers space for individual reasoning, question 1c positions students to verbally apply reasoning with their partner, and question 1d allows for students to apply their knowledge of both the process of this problem and their understanding of the material. Some students may have a preference toward one part of the question. Ensure equitable time for all four parts of the question, as each has its benefits for building deeper understanding.



Consider asking the following after question 1e:

- Why did you select the information you did for question 1e?
- Was it hard coming up with more examples? Why or why not?
- Was it hard to write a contradictory statement?
- Which statements were easier to write, the correct statements or the contradictory one? Why do you think that is?

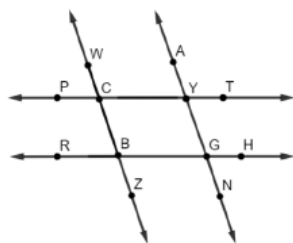
3.8.2 Collaboration: Parallel Lines (continued)

Work with your partner to correctly place the statements and reasons in the correct order by writing the letter.

Statement	Reason
1. G	1. P or T
2. C	2. P or T
3. A	3. N
4. D	4. R
5. H	5. K or S
6. F	6. L
7. B	7. K or S
8. E	8. M

3. Work with your partner to complete the proof in a different way.

Given: $\overline{PT} \parallel \overline{RH}$ and $\angle CBG \cong \angle CYG$
 Prove: $\overline{WZ} \parallel \overline{AN}$



Statement	Reason
1. $\overline{PT} \parallel \overline{RH}$	1. Given
2. $\angle CBG \cong \angle WCY$	2. If two parallel lines are intersected by a transversal, then corresponding angles are congruent.
3. $\angle CBG \cong \angle CYG$	3. Given
4. $\angle WCY \cong \angle CYG$	4. Transitive Property of Congruence
5. $\overline{WZ} \parallel \overline{AN}$	5. If two lines are intersected by a transversal so that alternate interior angles are congruent, then the lines are parallel.



Consider providing some proper placement of statements and/or reasons for the proof in question 2. Providing this extra scaffolding can offer a different entry point with students who are struggling with ordering the statements and reasons.



For students who may struggle to organize the statements and reasons, consider printing them out on a separate paper to be cut out to provide an alternative method for completing question 2.



While students are working on sorting the statements and reasons for question 2, monitor and make note of different orders the students place them in. Consider stopping students before moving on to question 3 to highlight different ordering, and how more than one order can be correct.



The proof in question 3 uses the same information as question 2. This was done intentionally to show that not only can the same statements be ordered differently, but also different statements in justifying a conclusion from a given may be taken all together. The goal is to show students there is more than one way to prove a relationship. The scaffolds in both question 2 and question 3 are purposeful to lead to two different proofs. There are a different number of steps and different statement and reasons in each.

3.8.3 Guided Instruction: Proofs Involving Angle Relationship

Component Overview: Students are guided through writing a flowchart proof and a paragraph proof given information related to parallel lines intersected by a transversal.

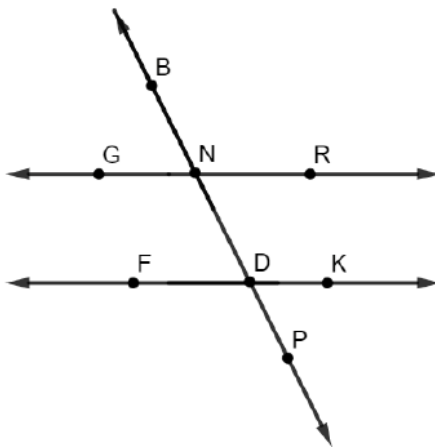


3.8.3 Guided Instruction: Proofs Involving Angle Relationship

- Complete the flowchart proof.

Given: $\angle RND$ and $\angle FDP$ are supplementary.

Prove: $\overleftrightarrow{GR} \parallel \overleftrightarrow{FK}$

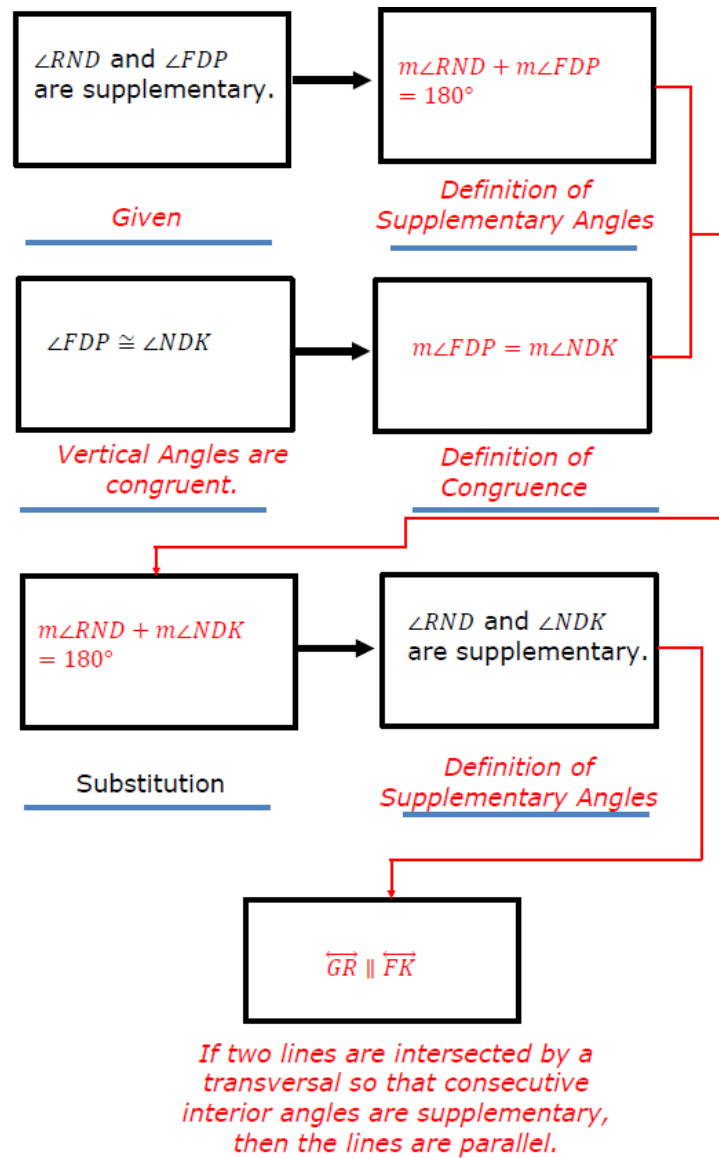


Students often find flowchart and paragraph proofs more challenging than the two-column proof. This component is designed to be guided so that teachers can help students become more familiar with these forms. Consider using think-aloud and questioning strategies while modeling the type of thinking used in completing these proofs.



Engage students in meaningful dialogue related to the similarities and differences between the different types of proofs. Encourage students to point out the benefit(s) of each form of proof.

3.8.3 Guided Instruction: Proofs Involving Angle Relationship (continued)



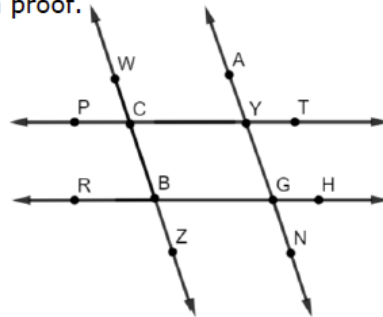
Guide students through the completion of this flowchart proof. Students should fill in the blank boxes with missing statements, and blank lines with missing reasons. Students should also draw arrows showing the connection of the statements, which is one of the benefits of a flow chart proof.

3.8.3 Guided Instruction: Proofs Involving Angle Relationship (continued)

2. Choose from the bank of statements and reasons to complete the paragraph proof.

Given: $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ and $\angle PCW \cong \angle HGN$

Prove: $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$



It is given that $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$.

Therefore, $\angle RBC \cong \angle PCW$ because if two parallel lines are intersected by a transversal then corresponding angles are congruent.

It is also given that $\angle PCW \cong \angle HGN$. By the Transitive Property of Congruence, it can be written that $\angle RBC \cong \angle HGN$. Therefore, it can be concluded that $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ because if two lines are intersected by a transversal so that alternate exterior angles are congruent then the lines are parallel.

Statements and Reasons

- | | |
|--|---|
| <ul style="list-style-type: none"> • $\overleftrightarrow{PT} \parallel \overleftrightarrow{RH}$ • $\overleftrightarrow{WZ} \parallel \overleftrightarrow{AN}$ • $\angle PCW \cong \angle HGN$ • $\angle ZBG \cong \angle HGN$ • $\angle GND$ • $\angle FDP$ | <ul style="list-style-type: none"> • Definition of Congruence • Transitive Property of Congruence • Definition of Supplementary Angles • Corresponding angles are congruent • Alternate exterior angles are congruent • Alternate interior angles are congruent |
|--|---|



Allow informal observation throughout the first part of this lesson to determine how “guided” this question should be in your classroom. This could mean small group instruction for some while others attempt independently, or it could mean teacher facilitation with students actively working as a whole group. The bank of statements and reasons provides a support for students, so you could also consider releasing the proof to partners to attempt on their own before going over with the whole class to clarify any questions or misconceptions.



Consider asking the following after completing question 2:

- If given a choice, why might you select one type of proof to complete over another type?
- Will this always be true? Why or why not?

3.8.4. Wrap-Up: Reflection

Component Overview: Students will complete a reflection by circling an icon that describes their understanding of each identified aspect of writing proofs.



3.8.4 Wrap-Up: Reflection

1. Circle the icon that best describes your understanding of each concept.

Answers will vary.



I need help.



I got it.



I could help someone.

Using <i>substitution</i> as a reason in a proof.	
Using the <i>transitive property of congruence</i> as a reason in a proof.	
Using the <i>definition of congruence</i> as a reason in a proof.	
Understanding what angles are congruent when two lines are parallel.	
Understanding what angles are supplementary when two lines are parallel.	



The information provided from the students here, coupled with what was observed during instruction, can provide valuable data for differentiating instruction. Recording this information and pairing it with assessment data provides insightful feedback for students' learning progressions.